PROJECT MARS
(Mathematical Reasoning Strategies)

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Study Background/Context

- Proportional reasoning exemplifies the kind of multiplicative thinking that is critical to mathematical development during the middle school years (Hiebert & Behr, 1988; National Research Council, 2001; National Mathematics Advisory Panel, 2008).
- Proportional reasoning is foundational for more advanced mathematics (e.g., algebra, geometry, trigonometry, and calculus) that students encounter in high school and college (NMAP, 2008).

Study Background/Context

- National and international assessments indicate that US students exhibit difficulties in the area of proportional reasoning.
- "…developing understanding of and applying proportional relationships…" in Grade 7 is one of four critical areas of focused instructional time (Common Core State Standards, 2010).

To reason with ratios and proportions

- Requires "understanding the multiplicative relationships between rational quantities" (Boyer, Levine, & Huttenlocher, 2008, p. 1478).

To solve word problems

- Students need to be able to recognize underlying mathematical structure (problem schema).
- Schemata are…

  "cognitive constructs that allow us to treat multiple elements of information in terms of larger higher-level units (or chunks) ... a schema can be described functionally by indicating how people with a specific level of acquisition of the schema are expected to act or make sense out of information in corresponding situations"(Kalyuga, 2006, p. 2).

To focus on the math structure

- Allows for determining what strategy to use for solving the problem, gives meaning to the operation/ concept involved in the problem, and helps transfer problem solving methods to problems with similar structures.
  - "*This is a rate problem*"
  - Rather than "*This is a bicycle problem*"

Our Model: Schema-Based Instruction

SBI draws upon schema theory and is designed to:

- Help students more effectively categorize problems by emphasizing the underlying mathematical problem structure; visual representations in the form of schematic representations are used to organize information in the problem (Marshall, 1995).
- Effectively integrate conceptual knowledge of principles in the specific domain with procedural knowledge (specific actions), foster flexibility in strategy use, and enhance metacognitive strategy knowledge (Jitendra et al., 2009).

SBI Framework

- Translate problem features into a coherent representation of the problem's mathematical structure using schematic diagrams.
- Apply a 4-step heuristic (DISC) that guides both problem translation and solution.
- Emphasize metacognitive strategy knowledge by having students reflect on the problem solving processes.
- Focus on multiple solution strategies and flexible application of those strategies.

Prior work with SBI

	Study 1	Study 2	Study 3
Students	148 (7th grade)	475 (7th grade)	1178 (7th grade)
Classrooms	8	21	42
Teachers	4	6	15
Design	Classrooms randomly assigned to condition		
Topics	Ratio, proportion	Ratio, proportion, percents	Ratio, proportion, percents
Effect size, Problem Solving test	0.45	0.75	1.24
Effect size, Retention test	0.56	0.49	1.27

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Project MARS' Objectives

- The primary objective is to assess the immediate and long-term impact of SBI on students' mathematical problem solving and proportional reasoning.
- Secondary objectives include assessing:
  - The immediate and long-term impact of SBI with seventh grade students when implemented by experienced SBI teachers.
  - Implementation factors that are associated with the effectiveness of SBI.
  - SBI and teacher-classroom characteristics that moderate the impact of student characteristics on mathematical problem solving and proportional reasoning.
  - Effects on students' metacognitive strategy knowledge as well as effects on student acceptability of SBI.

Study Participants

- In Studies 1 and 2 we will work with 100 seventh-grade teachers and their students throughout Minnesota.
  - These teachers will be randomly assigned to either Cohort 1 or Cohort 2. One classroom of each teacher's seventh-grade classes will participate in the study.
- Study 1 will examine the efficacy of SBI, while Study 2 will evaluate the sustainability of SBI.

Research Method

- A multi-wave, cluster, randomized-control design with random assignment of classrooms to treatment will be used to evaluate the efficacy of the SBI curriculum on student outcomes and the sustainability of SBI effectiveness.

Professional Development (PD)

Study 1 (AY 2012-2013)

- Cohort 1 teachers will receive two full days of PD conducted by project PI's in December 2012 and will implement the SBI curriculum in January/February 2013.
- Cohort 2 will continue to implement their district-adopted curriculum.

Study 2 (AY 2013-2014)

- Cohort 1 will continue to implement the SBI curriculum without additional PD.
- Cohort 2 will receive PD and implement the SBI curriculum.

Instructional Procedure and Materials

- The instructional approaches in Studies 1 and 2 will be used with all students in the classroom.
- This instruction will occur during typically scheduled mathematics instruction, 5 days a week for approximately 6 weeks.
- We will provide all of the testing and curriculum materials for the duration of the studies.

Student Assessment

- Students will complete a total of three assessments between January and June of each year. Each individual assessment will be administered before and immediately following the implementation of the SBI curriculum; in addition, one assessment will also be given at the end of the school year.
- On average, each of the assessments will be completed within a single math class (approximately 50 minutes).

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Teacher Assessment

- All teachers will complete a short teacher knowledge questionnaire related to the instructional content at the onset of the first year of participation in the study.
- We will conduct two observations in all classrooms to gather fidelity of treatment implementation.
- SBI teachers will participate in interviews to provide us with feedback on our materials.

Compensation

Cohort 1 teachers will receive:

$300 PD payment December 2012

$1,000 stipend for implementation of SBI curriculum May 2013

$1,000 stipend for implementation of SBI curriculum May 2014

$2,300 Total

Cohort 2 teachers will receive:

$400 stipend for implementation of district-adopted curriculum May 2013

$300 PD payment December 2013

$1,000 stipend for implementation of SBI curriculum May 2014

$1,700 Total

Project MARS will cover the cost of substitute teachers during the absence of participating teachers at PD. We can provide hotel accommodation for those participants who are traveling great distances; requests for overnight accommodations will be handled on a case-by-case basis.

Teacher Project Requirements

- Allow random assignment of teachers to Cohort 1 or 2.
- Commit to teach a "typical" seventh-grade math class for two years.
- Attend two full days of PD at U of M.
- Complete a test of teacher knowledge on related instructional content prior to implementing the SBI curriculum.
- Administer a battery of assessments to all participating students.
- Allow two video-recorded lesson observations.
- Complete an exit-interview at study completion.

Benefits of Participation

- Improve student learning of ratio, proportion, and percents.
- Collaborate with renowned researchers to implement a promising curricular intervention.
- Unique professional development opportunity to engage with other seventh-grade math teachers.
- Generous stipend paid to each participating teacher.
- Continuing Education Credits provided to those teachers implementing the SBI curriculum. As such, Cohort 1 will receive 40 CEU credits each year (80 total), and Cohort 2 will receive a total of 40 CEU credits (for year 2).

Project Personnel

Asha Jitendra, Rodney Wallace Professor, University of Minnesota – Principal Investigator

Jon Star, Nancy Pforzheimer Aronson Associate Professor, Harvard University – Co-Principal Investigator

Michael Harwell, Professor, University of Minnesota – Methodologist

Susan Slater – Project Coordinator

Danielle Dupuis and Amy Lein – Research Assistants



Thank you!

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